Bahae-Eddine Madir | PhD in Applied Mathematics

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Academic and Professional Background

University of Rouen Normandy – LMRS **Rouen, France**
Postdoctoral Researcher *Oct. 2026 – Present*
In collaboration with: Francky Luddens and Simon Bernard.
Research topic: Optimization of phase-change materials through inverse problems using scientific machine learning methods.

University of Rouen Normandy – LMRS **Rouen, France**
PhD in Applied Mathematics *Oct. 2022 – June 2026*
PhD supervisor: Ionut Danaila.
Research topic: Simulation of liquid–solid systems with phase change using physics-informed neural networks (PINNs).

University of Rouen Normandy **Rouen, France**
Master's Degree in Mathematical Modeling and Analysis *2020 – 2022*

Sidi Mohamed Ben Abdellah University, F.S.T. **Fez, Morocco**
Bachelor's Degree in Mathematics and Applications *2019 – 2020*

References

Bahae-Eddine Madir, Francky Luddens, Corentin Lothodé, and Ionut Danaila. Physics informed neural networks for heat conduction with phase change. *International Journal of Heat and Mass Transfer*, 252:127430, 2025.

Bahae-Eddine Madir and Antoine Lechevallier. Benchmarking pde problems via machine learning for enhanced computational modeling. *ESAIM: Proceedings and Surveys*, 81:33–54, 2025.

Teaching Activities

University of Rouen Normandy
Tutorials and Laboratory Sessions (96 h) *2025 – 2026*
Numerical Analysis (L3), Scientific Computing (L3), Algebra (L1).

University of Rouen Normandy
Tutorials and Laboratory Sessions (64 h) *2024 – 2025*
Numerical Analysis (L3), Scientific Computing (L3).

University of Rouen Normandy
Tutorials (64 h) *2023 – 2024*
Mathematical Analysis (L1).

Total teaching load: 224 hours

Scientific Activities and Conferences

PIML Day **Paris, France**
Methodological and Theoretical Advances in Physics-Informed ML *Apr. 2026*
Poster: *Physics-Informed Neural Networks for Liquid–Solid Systems with Phase Change.*

AI4Science – Summer School **Caen, France**
Advancing Scientific Discovery with AI *Jul. 2025*
Presentation: *Physics-Informed Neural Networks for Liquid–Solid Systems with Phase Change.*

CANUM 2024 **Île de Ré, France**
Numerical Analysis Conference *May 2024*
Presentation: *Physics-Informed Neural Networks for Heat Conduction with Phase Change.*

CEMRACS 2023 Hackathon **CIRM, Marseille, France**
Jul. – Aug. 2023
Research project: *Fourier Neural Operators for Incompressible Two-Phase Flow Simulation.*

CEMRACS 2023 **CIRM, Marseille, France**
Summer School in Scientific Machine Learning *Jul. 2023*

Skills

Languages: French (fluent), English (fluent).

Programming and Tools: Python (NumPy, SciPy, PyTorch, Optuna), C/C++, FreeFem++, MATLAB, ParaView, LaTeX, Git.

Numerical Methods: Finite differences, finite elements, PINNs.

Research Areas: Partial differential equations, scientific machine learning, fluid mechanics, heat transfer, phase change.